

$$i(j) = (\lambda, \mu); \quad \lambda = \{\alpha, \beta\}$$

$$x_{ij} = x_{ji}$$

$$x_{ij}(T) = x_{ij} \cdot f(E_i/T) (1 - f(E_j/T))$$

$$x_{ji}(T) = x_{ij} f(E_j/T) f(-E_i/T)$$

oder auch x_{ij}

$$\begin{aligned} \textcircled{1} \quad T_1 = \infty \rightarrow f(0) &= \Theta(E - V/2) \rightarrow x_{ij}(T_1) + x_{ji}(T_2) = \\ T_2 = 0 \rightarrow f(\infty) &= \frac{1}{2} \rightarrow x_{ij} \left[\Theta(E_i - \frac{V}{2}) \Theta(\frac{V}{2} - E_j) + \frac{1}{2} \cdot \frac{1}{2} \right] \end{aligned}$$



$$\left. \begin{aligned} R_{ij} &= x_{ij} \left[\frac{1}{4} + \frac{1}{4} \right] = \frac{5}{4} x_{ij} \\ R_{ji} &= x_{ij} \left[\frac{1}{4} + 0 \right] \end{aligned} \right\} \Rightarrow \dot{n} = 0 \Leftrightarrow (1-n_i) R_{ij} n_j = (1-n_j) R_{ji} n_i$$

$$\rightarrow \frac{n_i}{1-n_i} = \frac{1-n_j}{n_j} \left(\frac{1}{5} \right)$$

$$n_i + n_j = 1$$

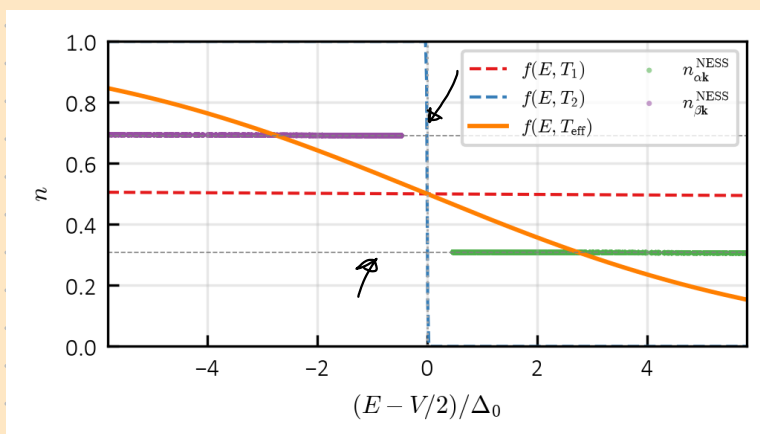
$$n_i = 1 - n_j \quad \left(\frac{n_j}{1-n_j} \right)^2 = 5 \Rightarrow$$

$$\begin{aligned} n_j &= \frac{\sqrt{5}}{1+\sqrt{5}} \quad , \quad n_i = \frac{1}{\sqrt{5}+1} \\ &\sim 0.65098 \quad \quad \quad \sim 0.3090 \end{aligned}$$

$$\rightarrow n_{\beta} - n_{\alpha} = \frac{\sqrt{5}-1}{\sqrt{5}+1}$$

$$\left. \begin{aligned} -1 \text{ gap} \\ \text{g. zu} \\ T_1 = \infty \\ T_2 = 0 \end{aligned} \right\} \rightarrow \frac{1}{V} \cdot \frac{\sqrt{5}+1}{\sqrt{5}-1} = \sum_k \frac{1}{N} \frac{1}{2\sqrt{z_k^2 + \Delta^2}}$$

$V_{\text{eff}} \sim 0.38 V$



BCS tok :
$$0 = 2\Gamma_{ij}^{++} (1-n_i)(1-n_j) - 2\Gamma_{ij}^{--} n_i n_j +$$

$$+ \Gamma_{ij}^{+-} (1-n_i)n_j - \Gamma_{ji}^{+-} (1-n_j)n_i = 0$$

$$\Gamma_{ij}^{++} = \chi_{ij} \left[\overbrace{\frac{1}{2} \cdot \frac{1}{2}}^{t_z=0} + \right]$$